

Ningning Xie

Professional Positions

- Aug 2023- **Assistant Professor**
University of Toronto, Toronto, Canada
- Nov 2022- **Research Scientist**
Google DeepMind, Toronto, Canada
- Oct 2021-2022 **Research Associate**
University of Cambridge, Cambridge, UK
- May-Sept 2021 **Research Intern**
DeepMind, London, UK
- Jan-Aug 2020 **Research Intern**
Microsoft Research, Redmond, WA
- Aug-Nov 2019 **Software Engineer Intern**
Google, New York City, NY

Education

- 2021 **Ph.D., Computer Science**
University of Hong Kong, Hong Kong
- 2015 **B.S. (with Honors), Computer Science and Technology**
Zhejiang University, Hangzhou, China

Awards and Honors

- 2025 ACM SIGPLAN Distinguished Paper Award (ICFP 2025)
- 2025 ACM SIGPLAN Distinguished Paper Award (PLDI 2025)
- 2023 Publication of the Year, University of Cambridge Hall of Fame Awards (PLDI 2023)
- 2021 ACM SIGPLAN Distinguished Paper Award (PLDI 2021)
- 2020 Rising Stars in EECS
- 2020 ACM SIGPLAN Distinguished Paper Award (POPL 2020)
- 2020 Special issue of ACM Transactions on Programming Languages and Systems (ESOP'18)

Publications

- OOPSLA'26 Andong Fan, Lionel Parreaux **Ningning Xie**. **Modular Type Safety for Traits with Extensible Variants and Deep Pattern Matching**. *41st ACM International Conference on Object-Oriented Programming, Systems, Languages, and Applications*.
- OOPSLA'26 Ka Wing Li, Maite Kramarz, **Ningning Xie**, Jeremy Yallop. **Mechanised Semantics of Multi-Stage Programming**. *41st ACM International Conference on Object-Oriented Programming, Systems, Languages, and Applications*.
- ESOP'26 Hasti Toossi, **Ningning Xie**. **Bidirectional Type Checking for Existential Types with Higher-Rank Polymorphism**. *35th European Symposium on Programming*.

- POPL'26 Matthew Toohey, Yanning Chen, Ara Jamalzadeh, **Ningning Xie**. **Extensible Data Types with Ad-Hoc Polymorphism**. *53rd ACM SIGPLAN Symposium on Principles of Programming Languages*.
- POPL'26 Michael Lee, **Ningning Xie**, Oleg Kiselyov, Jeremy Yallop. **Handling Scope Checks: A Comparative Framework for Dynamic Scope Extrusion Checks**. *53rd ACM SIGPLAN Symposium on Principles of Programming Languages*.
- ICFP'25 Tsung-Ju Chiang, **Ningning Xie**. **Multi-Stage Programming with Splice Variables**. *30th International Conference on Functional Programming*.
ACM SIGPLAN Distinguished Paper Award
- PLDI'25 Andong Fan, Han Xu, **Ningning Xie**. **Practical Type Inference with Levels**. *46th Conference on Programming Language Design and Implementation*.
ACM SIGPLAN Distinguished Paper Award
- PLDI'25 Gordon Plotkin, **Ningning Xie**. **Handling the Selection Monad**. *46th Conference on Programming Language Design and Implementation*.
- ICFP'24 Tsung-Ju Chiang, Jeremy Yallop, Leo White, **Ningning Xie**. **Staged Compilation with Module Functors**. *29th International Conference on Functional Programming*.
- ICFP'24 **Ningning Xie***, Daniel D. Johnson*, Dougal Maclaurin, Adam Paszke. **Parallel Algebraic Effect Handlers**. *29th International Conference on Functional Programming*.
- ICFP'23 **Ningning Xie**, Leo White, Olivier Nicole, Jeremy Yallop. **MacoCam!: Staging Composable and Compilable Macros**. *28th International Conference on Functional Programming*.
- TyDe'23 Adam Paszke, **Ningning Xie** **Infix-extensible Record Types for Tabular Data**. *8th ACM SIGPLAN International Workshop on Type-Driven Development*.
- PLDI'23 Jeremy Yallop, **Ningning Xie**, Neel Krishnaswami. **flap: A Deterministic Parser with Fused Lexing**. *44th Conference on Programming Language Design and Implementation*.
Publication of the Year, University of Cambridge Hall of Fame Awards
- APLAS'22 Xu Xue, Bruno C. d. S. Oliveira, **Ningning Xie**. **Applicative Intersection Types**. *20th Asian Symposium on Programming Languages and Systems*.
- OOPSLA'22 **Ningning Xie**, Youyou Cong, Kazuki, Ikemori, Daan Leijen. **First-class Names for Effect Handlers**. *37th ACM International Conference on Object-Oriented Programming, Systems, Languages, and Applications*.
- ECOOP'22 Baber Rehman, Xuejing Huang, **Ningning Xie**, Bruno C. d. S. Oliveira. **Union Types with Disjoint Switches**. *36th European Conference on Object Oriented Programming*.
- MLSys'22 **Ningning Xie**, Tamara Norman, Diminik Grewe, Dimitrios Vytiniotis. **Synthesizing Optimal Parallelism Placement and Reduction Strategies on Hierarchical Systems for Deep Learning**. *5th Conference on Machine Learning and Systems*.
- POPL'22 **Ningning Xie**, Matthew Pickering, Andres Löb, Nicolas Wu, Jeremy Yallop, Meng Wang. **Staging with Class: A Specification for Typed Template Haskell**. *49th Symposium on Principles of Programming Languages*.
- ICFP'21 **Ningning Xie**, Daan Leijen. **Generalized Evidence Passing for Effect Handlers: Efficient Compilation of Effect Handlers to C**. *26th International Conference on Functional Programming*.
- PLDI'21 Alex Reinking*, **Ningning Xie***, Leonardo de Moura, Daan Leijen. **Perceus: Garbage Free Reference Counting with Reuse**. *42nd Conference on Programming Language Design and Implementation*.
ACM SIGPLAN Distinguished Paper Award
- Haskell'20 **Ningning Xie**, Daan Leijen. **Effect Handlers in Haskell, Evidently**. *13th Haskell Symposium*.

- ICFP'20 **Ningning Xie**, Jonathan Brachthäuser, Daniel Hillerström, Philipp Schuster, Daan Leijen. *Effect Handlers, Evidently*. *25th International Conference on Functional Programming*.
- POPL'20 **Ningning Xie**, Richard A. Eisenberg, Bruno C. d. S. Oliveira. *Kind Inference for Datatypes*. *47th Symposium on Principles of Programming Languages*.
ACM SIGPLAN Distinguished Paper Award
- ECOOP'20 **Ningning Xie**, Bruno C. d. S. Oliveira, Xuan Bi, Tom Schrijvers. *Row and Bounded Polymorphism via Disjoint Polymorphism*. *34th European Conference on Object Oriented Programming*, ECOOP 2020.
- TOPLAS'20 **Ningning Xie**, Xuan Bi, Bruno C. d. S. Oliveira, Tom Schrijvers. *Consistent Subtyping for All*. *ACM Transaction on Programming Languages and Systems*, TOPLAS 2020.
- ICFP'19 Gert-Jan Bottu, **Ningning Xie**, Koar Marntirosian, Tom Schrijvers. *Coherence of Type Class Resolution*. *24th International Conference on Functional Programming*.
- ESOP'19 Xuan Bi, **Ningning Xie**, Bruno C. d. S. Oliveira, Tom Schrijvers. *Distributive Disjoint Polymorphism for Compositional Programming*. *28th European Symposium on Programming*, ESOP 2019.
- ESOP'18 **Ningning Xie**, Xuan Bi, Bruno C. d. S. Oliveira. *Consistent Subtyping for All*. *27th European Symposium on Programming*.
Selected for Special Issue of ACM Transaction on Programming Languages and Systems
- ESOP'18 **Ningning Xie**, Bruno C. d. S. Oliveira. *Let Arguments Go First*. *27th European Symposium on Programming*, ESOP 2018.

Professional Activities

Program Chair

- Haskell 2026 19th ACM SIGPLAN International Symposium on Haskell
- PEPM 2026 28th ACM SIGPLAN Workshop on Partial Evaluation and Program Manipulation
- Haskell 2025 18th ACM SIGPLAN International Symposium on Haskell
- HIW 2021 13th Haskell Implementors' Workshop

Organizer

- POPL 2026 Publicity Chair.
- ICFP 2025 Programming Languages Mentoring Workshop (PLMW) Co-chair.
- PLDI 2025 Artifact Evaluation Co-Chair.
- POPL 2025 Publicity Chair.
- ICFP 2024 Programming Languages Mentoring Workshop (PLMW) Co-chair.
- PLDI 2024 Artifact Evaluation Co-Chair.

Program Committee

- 2027 POPL
- 2026 LOPSTR+PPDP. ML.
- 2025 ASPLOS. PLDI. ICFP.
- 2024 TACAS. WITS. POPL SRC. OCaml.
- 2023 POPL. PLDI. PADL. TFP. TyDe.
- 2022 ICFP. TFP. FLOPS. Scala. ML. HOPE. SPLASH SRC.
- 2021 PADL
- 2019 Haskell. HIW.

Steering Committee

2025-2029 ICFP Steering Committee (member at large)
2025-2026 PLMW Steering Committee
2020-2023, Haskell Symposium Steering Committee
2025-2028

Teaching

- CSC2126: Topics in Programming Languages
Fall 2024, Fall 2025, Fall 2026
- CSC324: Principles of Programming Languages
Winter 2024, Fall 2024, Fall 2025, Fall 2026
- CSC2125: Types and Programming Languages
Fall 2023